



BAULKHAM HILLS HIGH SCHOOL

2010

YEAR 11 YEARLY EXAMINATIONS

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- All necessary working should be shown in every question
- Marks may be deducted for careless or badly arranged work

Total marks – 84

- Attempt Questions 1 – 7
- All questions are of equal value

Total marks – 84

Attempt Questions 1 – 7

All questions are of equal value

Answer each question on a SEPARATE piece of paper clearly marked Question 1, Question 2, etc. Each piece of paper must show your name.

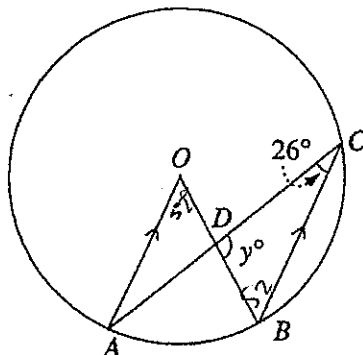
Question 1 (12 marks) Use a *separate* piece of paper

Marks

- a) (i) Expand $\left(x + \frac{1}{x}\right)^2$ 1
- (ii) Suppose that $x + \frac{1}{x} = 3$, evaluate $x^2 + \frac{1}{x^2}$ without finding the value of x . 2
- b) Solve the inequality $\frac{x+1}{x-1} \geq 3$ 3
- c) Find the coordinates of the point which divides the interval joining the points $(-6, 2)$ and $(4, 7)$ in the ratio 3:2 2
- d) Find the exact value of $\sin 15^\circ$ by using the expansion for $\sin(\alpha + \beta)$ 2
- e) In how many different ways can people between the ages of 20 and 40 (inclusive), be classified by sex, age and political affiliation? 2
(Assume that the political affiliates are Labor, Liberal, Greens and Independent)

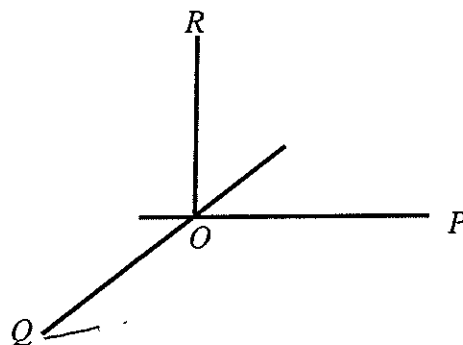
Question 2 (12 marks) Use a *separate* piece of paper

- a) The variable point $(5t, 4t^2)$ lies on a parabola. Find the cartesian equation for this parabola. 2
- b) The points A, B and C lie on a circle with centre O .
The lines AO and BC are parallel, and OB and AC intersect at D .
Also $\angle ACB = 26^\circ$ and $\angle BDC = y^\circ$, as shown in the diagram.



- (i) State why $\angle AOB = 52^\circ$ 1
- (ii) Find y . Justify your answer. 2
- c) A, B and C are the points $(-3, -2)$, $(-4, 2)$ and $(2, 5)$ respectively. Find the size of $\angle ABC$, correct to the nearest degree. 3

d)



In the diagram, the points O, P and Q are in the same plane. R is a point vertically above O . P and Q are 750 metres apart and $\angle POQ = 120^\circ$, $\angle QRO = 30^\circ$ and $\angle PRO = 60^\circ$.

- (i) Redraw the diagram on your answer sheet, labelling all of the given information. 1
- (ii) Find the height of R above O , correct to the nearest metre. 3

Question 3 (12 marks) Use a *separate* piece of paper

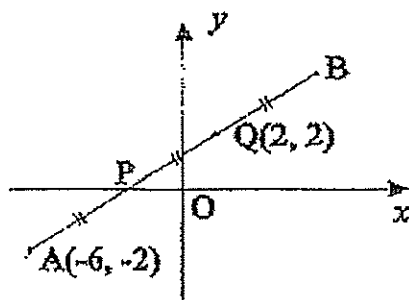
- a) Write down the equation of the vertical asymptote of $y = \frac{2x}{x+6}$ 1
- b) If $\sin \theta = \frac{3}{7}$ and $90^\circ < \theta < 180^\circ$, find the exact value of;
- (i) $\sin 2\theta$ 2
- (ii) $\tan \frac{\theta}{2}$ 3
- c) Solve for θ , correct to the nearest degree where necessary, where $0^\circ \leq \theta \leq 360^\circ$
- (i) $2 \cos 2\theta = 4 \cos \theta - 3$ 3
- (ii) $\cos \theta - \sqrt{3} \sin \theta = 2$ 3

Question 4 (12 marks) Use a *separate* piece of paper

- a) In how many ways can the letters of the word **PARRAMATTA** be arranged if;
- (i) all the letters are used? 2
- (ii) all the letters are used and the word begins and ends with the letter **A**? 2
- b) There are nine people applying for four jobs. How many different ways can the four jobs be allocated if;
- (i) the jobs are allocated randomly? 2
- (ii) out of the nine people, five have their HSC and three of the jobs require the HSC? 2
- c) There are two rows of chairs, with three in the first row and four in the second row. How many ways can seven people be seated if;
- (i) Daniel and Joshua must sit in the second row? 2
- (ii) Bryan will not sit in the same row as Shirley and Harleen must sit in the first row? 2

Question 5 (12 marks) Use a *separate* piece of paper

- a) Solve $\tan x \tan 2x = 1$ where $0^\circ \leq x \leq 360^\circ$ 3
- b) The diagram shows the line interval AB **trisected** at P and Q . 3
The coordinates of A are $(-6, -2)$ and of Q are $(2, 2)$



Find the coordinates of B using two different methods.

- c) $P(2ap, ap^2)$ is a point on the parabola $x^2 = 4ay$, which has focus S .
- (i) Derive the equation of the tangent at P . 2
- (ii) Hence show that the tangent meets the x -axis at the point $T(ap, 0)$ 1
- (iii) Find the coordinates of M , the point that divides ST externally in the ratio 2:1 2
- (iv) Explain why $PS = PM$. 1

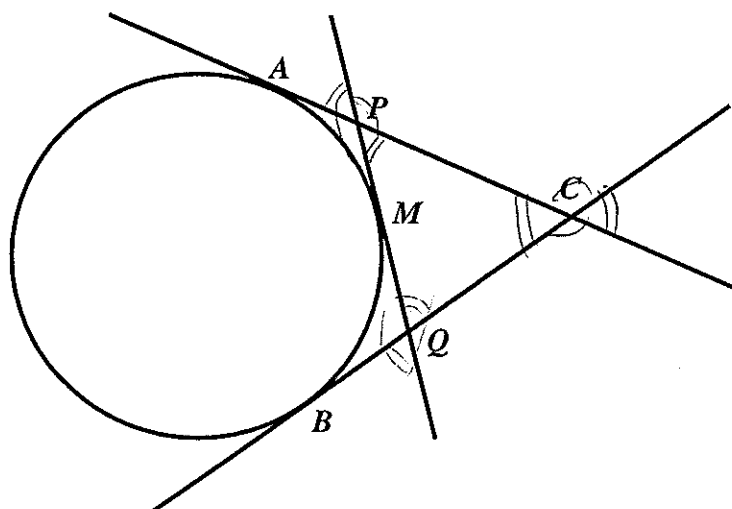
Question 6 (12 marks) Use a *separate* piece of paper

- a) There are seven points on a plane so that no three points lie on the same straight line. How many quadrilaterals can be formed if the quadrilateral must contain point A? 1

- b) (i) Prove that $\cot \theta - 2 \cot 2\theta \equiv \tan \theta$ 1

- (ii) Hence deduce that $\tan x + 2 \tan 2x + 4 \tan 4x \equiv \cot x - 8 \cot 8x$ 2

- c) 3



A and B are two points on a circle. Tangents at A and B meet at C. A third tangent cuts CA and CB at P and Q respectively, as shown in the diagram.

Show that the perimeter of $\triangle CPQ$ is independent of PQ.

- d) The word **EQUATIONS** contains all five vowels. How many seven letter words, including nonsense words, consisting of all five vowels can be formed from the letters of **EQUATIONS**? 2

- e) How many ways can six men and four women be seated around a table, if a particular husband and wife **do not** want to sit next to each other? 3

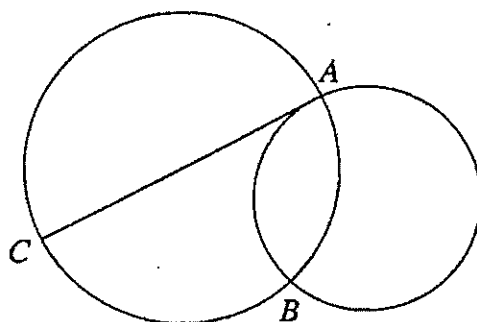
Question 7 (12 marks) Use a *separate* piece of paper

Marks

a) If $\sqrt{5} = 2 + \frac{1}{a}$, show that $a = 4 + \frac{1}{a}$

2

- b) The diagram shows two circles intersecting at A and B . The diameter of one circle is AC .



- (i) Copy the diagram onto your answer sheet and draw a straight line through A , parallel to CB , meeting the second circle at D . 1
- (ii) Prove that BD is a diameter of the second circle. 2
- (iii) Suppose that BD is parallel to CA . Prove that the circles have equal radii. 2
- c) The three sides of a triangle have lengths $p - q$, $p + q$ and p , where $p > q > 0$. The largest and smallest angles of the triangle are α and β respectively.
- (i) Show, using the cosine rule, that $4(1 - \cos \alpha)(1 - \cos \beta) = \cos \alpha + \cos \beta$ 3
- (ii) In the case that $\alpha = 2\beta$, show that $\cos \beta = \frac{3}{4}$ 2

END OF EXAMINATION

Question 1 (12)

a) (i) $(x + \frac{1}{x})^2 = x^2 + 2 + \frac{1}{x^2}$ (1)

(ii) $x^2 + \frac{1}{x^2} = (x + \frac{1}{x})^2 - 2$
 $= 3^2 - 2$ -1
 $= 7$ -1 (2)

b) $\frac{x+1}{x-1} \geq 3$

$x-1 \neq 0$ $x+1 = 3x-3$
 $x \neq 1$ $2x = 4$ critical pts -1
 $x = 2$ evidence of testing -1

$\frac{1}{x} \quad 1 \quad \checkmark \quad 2 \quad x$
 $1 < x \leq 2$ -1 (3)

c) $(-6, 2) \quad (4, 7)$
 $3:2$

$(\frac{-12+12}{5}, \frac{4+21}{5})$ evidence of correct method -1
 $= (0, 5)$ -1 (2)

d) $\sin 15^\circ = \sin(45-30)$
 $= \sin 45 \cos 30 - \cos 45 \sin 30$ -1
 $= (\frac{1}{\sqrt{2}})(\frac{\sqrt{3}}{2}) - (\frac{1}{\sqrt{2}})(\frac{1}{2})$
 $= \frac{\sqrt{3}-1}{2\sqrt{2}}$ -1 (2)

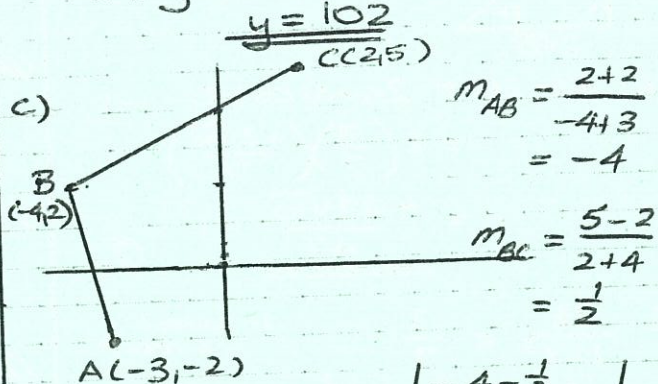
e) Ways = $21 \times 2 \times 4$ evidence of correct method -1
 $= 168$ -1 (2)

Question 2 (12)

a) $x = 5t$ $y = 4t^2$ attempt to eliminate parameter -1
 $t = \frac{x}{5}$ $= 4(\frac{x^2}{25})$
 $y = \frac{4x^2}{25}$ -1 (2)

b) (i) $\angle$ at centre twice $\angle$ at circumference standing on same arc. (1)

(ii) $\angle OAC = 26^\circ$ (alternate $\angle$'s, $AO \parallel BC$)
 $\angle ADO = y$ (vertically opposite $\angle$'s =)
 $\angle ACD = 52^\circ$ (proven in (i))
 $\angle OAC + \angle ADO + \angle ACD = 180$ ($\angle$ sum $\triangle ADO$)
 $26 + y + 52 = 180$ 1- solution
 $y = 102$ 1- reasoning (2)

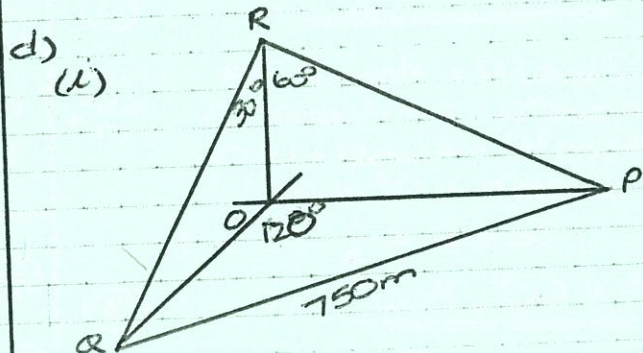


$\tan \angle ABC = \left| \frac{-4 - \frac{1}{2}}{1 + (-4)(\frac{1}{2})} \right|$
 $= \left| \frac{-\frac{9}{2}}{-1} \right|$ 1- substituting correct signs into correct formula
 $= \frac{9}{2}$

$\angle ABC = 77^\circ$ -1

however $\angle ABC$ is obtuse

$\therefore \angle ABC = 103^\circ$ -1 (3)



(ii) $\frac{QO}{OR} = \tan 30^\circ$ $\frac{PO}{OR} = \tan 60^\circ$ 1- expressing QO, PO in terms of OR.
 $QO = \frac{OR}{\sqrt{3}}$ $PO = \sqrt{3} OR$

$PQ^2 = OQ^2 + OP^2 - 2 \cdot OQ \cdot OP \cos 120^\circ$
 $750^2 = \frac{1}{3} OR^2 + 3 OR^2 - 2 OR^2 (-\frac{1}{2})$

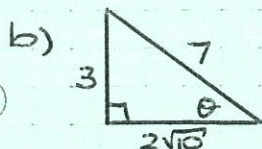
$750^2 = \frac{13}{3} OR^2$ 1- attempting to use Cosine Rule.
 $OR^2 = \frac{3 \times 750^2}{13}$

$OR = 360.2883461 \dots$ -1

$OR = 360 \text{ m}$ (to nearest m) (3)

Question 3 (12)

a) $x = -6$ (1)



(i) $\sin 2\theta = 2 \sin \theta \cos \theta$ -1
 $= 2 \left(\frac{3}{7} \right) \left(\frac{-2\sqrt{10}}{7} \right)$
 $= -\frac{12\sqrt{10}}{49}$ -1 (2)

(ii) $\frac{2t}{1+t^2} = \frac{3}{7}$ 1- correct t result
 $14t = 3 + 3t^2$
 $3t^2 - 14t + 3 = 0$
 $t = \frac{14 \pm \sqrt{160}}{6}$ -1
 $= \frac{14 \pm 4\sqrt{10}}{6}$
 $= \frac{7 \pm 2\sqrt{10}}{3}$

however $\frac{0}{2}$ is acute

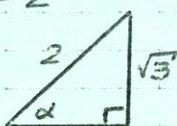
$\therefore \tan \frac{\theta}{2} = \frac{7+2\sqrt{10}}{3}$ (3)

c) $2 \cos 2\theta = 4 \cos \theta - 3$ 1- using $\cos 2\theta$ result
 $4 \cos^2 \theta - 2 = 4 \cos \theta - 3$
 $4 \cos^2 \theta - 4 \cos \theta + 1 = 0$
 $(2 \cos \theta - 1)^2 = 0$

$\cos \theta = \frac{1}{2}$ -1

$\theta = 60^\circ, 300^\circ$ -1 (3)

(ii) $\cos \theta - \sqrt{3} \sin \theta = 2$



$2 \cos(\theta + 60) = 2$

$\cos(\theta + 60) = 1$

$\theta + 60 = 0, 360$ -1 (both required)

$\theta = -60, 300$

$\therefore \theta = 300^\circ$ -1 (3)

OR $\frac{1-t^2}{1+t^2} - \frac{2\sqrt{3}t}{1+t^2} = 2$ 1- correct t results
 $1-t^2 - 2\sqrt{3}t = 2 + 2t^2$
 $3t^2 + 2\sqrt{3}t + 1 = 0$

$(\sqrt{3}t + 1)^2 = 0$

$t = -\frac{1}{\sqrt{3}}$ -1

$\frac{\theta}{2} = 150^\circ$

$\theta = 300^\circ$ -1

Question 4 (12)

Note: final answer not required for mark.

a) (i) Ways = $\frac{10!}{4!2!2!}$

$= 37800$

(ii) Ways = $1 \times 1 \times \frac{8!}{2!2!2!}$

$= 5040$ (2)

b) (i) Ways = 9C_4

$= 126$

(ii) Ways = ${}^5C_3 \times {}^6C_2$

$= 60$ (2)

c) (i) Ways = ${}^4P_2 \times 5!$

$= 1440$ (2)

(ii) Ways = ${}^3P_1 \times 2 \times 4 \times 2 \times 4!$

$= 1152$ (2)

NOTE: If student assumes (i) conditions.

Ways = ${}^4P_2 \times {}^3P_1 \times 2 \times 2 \times 2 \times 2!$
 $= 576$

Question 5 (12)

a) $\tan x \tan 2x = 1$

let $t = \tan x$

$t \left(\frac{2t}{1-t^2} \right) = 1$

$2t^2 = 1 - t^2$

$3t^2 = 1$

$t^2 = \frac{1}{3}$ -1

$t = \pm \frac{1}{\sqrt{3}}$

$\tan x = \pm \frac{1}{\sqrt{3}}$

$x = 30^\circ, 150^\circ, 210^\circ, 330^\circ$ -1 (3)

b) B divides AQ externally in the ratio 3:1

$$A(-6, -2) \quad Q(2, 2)$$

X
- 3:1

$$B\left(\frac{-6-3}{-2}, \frac{-2-3}{-2}\right)$$

$$= (6, 4)$$

Q divides AB internally in the ratio 2:1

$$A(-6, -2) \quad B(x, y)$$

X
2:1

$$Q\left(\frac{-6+2x}{3}, \frac{-2+2y}{3}\right)$$

$$\frac{-6+2x}{3} = 2 \quad \frac{-2+2y}{3} = 2$$

$$-6+2x = 6 \quad -2+2y = 6$$

$$2x = 12 \quad 2y = 8$$

$$x = 6 \quad y = 4$$

$$\therefore B(6, 4)$$

P is midpoint of AQ

$$P\left(\frac{-6+2}{2}, \frac{-2+2}{2}\right) = (-2, 0)$$

Q is midpoint of PB

$$Q\left(\frac{-2+x}{2}, \frac{0+y}{2}\right)$$

$$\frac{-2+x}{2} = 2 \quad \frac{y}{2} = 2$$

$$-2+x = 4 \quad y = 4$$

$$x = 6$$

$$\therefore B(6, 4)$$

A to Q = 2 parts

$$x: 2+6=8 \therefore 1 \text{ part} = 4$$

$$y: 2+2=4 \therefore 1 \text{ part} = 2$$

A to B = 3 parts

$$x = -6 + 3 \times 4 = 6$$

$$y = -2 + 3 \times 2 = 4$$

$$\therefore B(6, 4)$$

(3)

1- progress towards one correct solution

1- one correct solution

1- another different correct solution must use different method

$$c) (i) \quad y = \frac{x^2}{4a}$$

$$\frac{dy}{dx} = \frac{2x}{4a}$$

$$= \frac{x}{2a}$$

when $x = 2ap$, $\frac{dy}{dx} = p$ -1
 $\therefore$ slope of tangent is p

$$y - ap^2 = p(x - 2ap)$$

$$y - ap^2 = px - 2ap^2$$

$$y = px - ap^2 \quad -1 \quad (2)$$

(ii) x intercept occurs when $y = 0$

$$\text{i.e. } 0 = px - ap^2$$

$$px = ap^2$$

$$x = ap$$

$$\therefore T(ap, 0)$$

Note! if tangent in (i) incorrect T must be correct for their equation!

(1)

(iii)

$$S(0, a) \quad T(ap, 0)$$

X
-2:1

$$M\left(\frac{-2ap}{-2}, \frac{a}{-2}\right)$$

1- progress towards solution

(2)

$$= (2ap, -a) \quad -1$$

(iv) M lies on the directrix, by the definition of a parabola any point on the parabola is equidistant to the focus and directrix. (1)

Question 6 (12)

$$a) \text{ Quadrilaterals} = {}^6C_3$$

(1)

$$= 20$$

$$b) (i) \cot \theta - 2 \cot 2\theta$$

$$= \frac{1}{\tan \theta} - \frac{2(1 - \tan^2 \theta)}{2 \tan \theta}$$

$$= \frac{1 - 1 + \tan^2 \theta}{\tan \theta}$$

$$= \frac{\tan^2 \theta}{\tan \theta}$$

$$= \tan \theta$$

(1)

$$(iii) \tan x + 2 \tan 2x + 4 \tan 4x$$

$$= \cot x - 2 \cot 2x + 2 \cot 2x - 4 \cot 4x + 4 \cot 4x - 8 \cot 8x$$

$$= \underline{\cot x - 8 \cot 8x} \quad (2)$$

1- for showing use of (i)

$$c) \text{ Perimeter} = CP + CQ + PQ$$

$$AP = PM \quad (\text{tangents from external pt} =)$$

$$BQ = QM \quad (\quad " \quad)$$

$$PQ = PM + MQ \quad (\text{common side})$$

$$\therefore PQ = AP + BQ$$

$$\text{Perimeter} = CP + AP + CQ + BQ = AC + BC \quad (3)$$

$$\underline{\text{which is independent of PQ}}$$

2- solution 1- reasoning

$$d) \text{ Words} = 1 \times {}^4C_2 \times 7! \quad \begin{matrix} \text{1- selecting letters} \\ \text{1- arranging letters} \end{matrix}$$

$$= \underline{30240} \quad (2)$$

$$e) \text{ Total Ways}$$

1- placing H and W in circle

$$= 1 \times 7 \times 8!$$

$$= \underline{282240}$$

1- arranging other 8 is 8!

1- final answer

ie decided 1 if 7! or similar used.

OR

$$\text{Total, no restrictions} = 9!$$

$$\text{Total, HW together} \quad \begin{matrix} \text{1- demonstrate circle (n-1)!} \end{matrix}$$

$$= 2! \times 8!$$

1- ways together

$$\text{Total, HW no together}$$

$$= 9! - 2! \times 8!$$

-1

$$= \underline{282240}$$

Question 7 (12)

$$a) \sqrt{5} = 2 + \frac{1}{a}$$

$$\frac{1}{a} = \sqrt{5} - 2$$

$$a = \frac{1}{\sqrt{5}-2} \times \frac{\sqrt{5}+2}{\sqrt{5}+2}$$

$$= \sqrt{5} + 2$$

$$4 + \frac{1}{a} = 4 + \sqrt{5} - 2$$

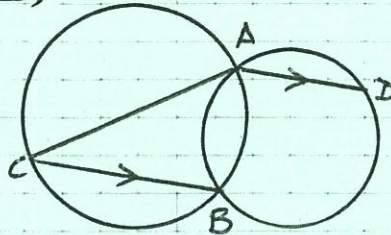
$$= \sqrt{5} + 2$$

$$= \underline{a}$$

1- progress towards solution

(2)

b)



Note: BC does not have to be drawn in.

(1)

$$(ii) \angle ABC = 90^\circ \quad (\angle \text{ in semicircle})$$

$$\angle ABC = \angle BAD \quad (\text{alternate } \angle s = AD \parallel CB)$$

$$\therefore \angle BAD = 90^\circ$$

Thus BD is diameter ($\angle$ subtended at circumference = 90°)

1- solution

1- reasoning

(2)

$$(iii) AD \parallel CB \quad (\text{given})$$

$$BD \parallel CA \quad (\text{given})$$

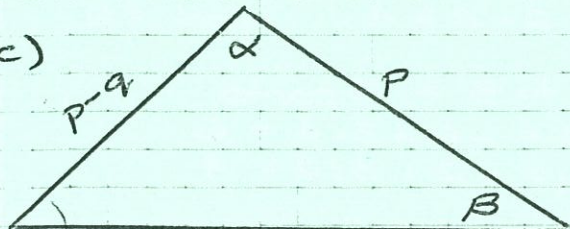
$\therefore$ ADBC is a ||gram (opposite sides ||)

$$AC = BD \quad (\text{opposite sides in ||gram})$$

If diameters are equal, circles must have equal radii. (2)

1- solution, 1 reasoning

Note: must prove ||gram, not just assume.



$$\cos \alpha = \frac{p^2 + (p-q)^2 - (p+q)^2}{2p(p-q)}$$

$$= \frac{p^2 - 4pq}{2p(p-q)}$$

$$= \frac{p - 4q}{2(p-q)}$$

Similarly

$$\cos \beta = \frac{p + 4q}{2(p+q)}$$

1- using cosine rule to find $\cos \alpha$ or $\cos \beta$

$$\begin{aligned}
 & 4(1 - \cos \alpha)(1 - \cos \beta) \\
 &= 4\left(1 - \frac{p-4q}{2(p-q)}\right)\left(1 - \frac{p+4q}{2(p+q)}\right) \\
 &= 4 \times \frac{p+2q}{2(p-q)} \times \frac{p-2q}{2(p+q)} \\
 &= \frac{p^2 - 4q^2}{p^2 - q^2}
 \end{aligned}$$

$$\begin{aligned}
 & \cos \alpha + \cos \beta \\
 &= \frac{p-4q}{2(p-q)} + \frac{p+4q}{2(p+q)} \\
 &= \frac{p^2 + pq - 4pq - 4q^2 + p^2 - pq + 4pq - 4q^2}{2(p-q)(p+q)} \\
 &= \frac{2p^2 - 8q^2}{2(p-q)(p+q)} \\
 &= \frac{p^2 - 4q^2}{p^2 - q^2}
 \end{aligned}$$

$$= \underline{4(1 - \cos \alpha)(1 - \cos \beta)} \quad (3)$$

1 - substituting
for $\cos \alpha$ and
 $\cos \beta$

1 - $\frac{p^2 - 4q^2}{p^2 - q^2}$ in
either LHS or RHS.

1 - $\frac{p^2 - 4q^2}{p^2 - q^2}$ in both.

$$(ii) \alpha = 2\beta$$

$$4(1 - \cos 2\beta)(1 - \cos \beta)$$

$$= \cos 2\beta + \cos \beta$$

$$\begin{aligned}
 & 4(2 - 2\cos^2 \beta)(1 - \cos \beta) \\
 &= 2\cos^2 \beta - 1 + \cos \beta
 \end{aligned}$$

$$\begin{aligned}
 & 8(1 - \cos^2 \beta)(1 - \cos \beta) \\
 &= 2\cos^2 \beta + \cos \beta - 1
 \end{aligned}$$

$$\begin{aligned}
 & 8(1 - \cos \beta)^2(1 + \cos \beta) \\
 &= (2\cos \beta - 1)(\cos \beta + 1)
 \end{aligned}$$

$$8(1 - \cos \beta)^2 = 2\cos \beta - 1$$

(Note: $\cos \beta \neq -1$)

$$8 - 16\cos \beta + 8\cos^2 \beta = 2\cos \beta - 1$$

$$8\cos^2 \beta - 18\cos \beta + 9 = 0$$

$$(4\cos \beta - 3)(2\cos \beta + 3) = 0$$

$$\cos \beta = \frac{3}{4} \text{ or } \cos \beta = -\frac{3}{2}$$

however β is acute

$$\therefore \underline{\underline{\cos \beta = \frac{3}{4}}} \quad (2)$$

1 - progress towards
correct solution
including at least
correct use of
 $\cos 2\alpha$ substitution

1 - correct solution.